SENTHILKUMARAN P

Generative AI Application Developer | AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI & Data Science graduate with hands-on experience designing and developing AI-powered applications using Generative AI, Large Language Models (LLMs), Machine Learning, and Computer Vision. Experienced in building end-to-end AI solutions, including multi-agent systems, Retrieval-Augmented Generation (RAG) pipelines, REST APIs, and intelligent automation using Python, LangChain, LangGraph, CrewAI, Google Gemini, TensorFlow, and FastAPI. Proven ability to deliver production-ready projects through internships, hackathons, and research initiatives while applying strong software engineering practices, data-driven problem solving, and collaborative development. Passionate about building scalable AI applications that solve real-world business challenges and continuously exploring emerging AI technologies.

TECHNICAL SKILLS

Category	Skills
Programming	Python • SQL
Generative AI	LLMs • RAG • AI Agents • LangChain • LangGraph • CrewAI • MCP
Machine Learning	TensorFlow • PyTorch • Scikit-learn • CNNs • Transfer Learning
Computer Vision	YOLOv8 • OpenCV • MediaPipe • SDXL • ControlNet
Backend	FastAPI • Flask • REST APIs
Data Science	EDA • Feature Engineering • Statistical Analysis • Data Visualization (Matplotlib, Seaborn)
Databases & Tools	MySQL • ChromaDB • Git • GitHub • Microsoft Excel • CLI

PROFESSIONAL EXPERIENCE

AI Developer Intern

Python and Ladder (Uraikathai) Jan 2026 – Present

- **Engineered** an AI-powered interactive storytelling pipeline by integrating SDXL, img2img, and ControlNet to generate context-aware, visually consistent narrative assets, enhancing end-to-end Generative AI application workflows.
- **Researched and optimized** character consistency, scene coherence, and prompt orchestration through iterative experimentation, parameter tuning, and human-in-the-loop evaluation, improving the quality and reliability of generative outputs.
- **Designed** reproducible AI experimentation workflows by maintaining structured research documentation, version-controlled artifacts, and evaluation metrics, enabling systematic model validation and continuous improvement.

AI & ML Intern

Apex Seekers Edtech Private Limited *Jul 2025 – Aug 2025*

- **Designed and implemented** end-to-end machine learning pipelines using Python, Pandas, and Scikit-learn, covering data preprocessing, feature engineering, model training, hyperparameter optimization, and performance evaluation.
- **Applied** advanced data preprocessing techniques, including data cleaning, normalization, feature selection, and transformation, improving model robustness and predictive performance across multiple supervised learning tasks.
- **Conducted** comparative model evaluation using appropriate performance metrics and analytical validation techniques, identifying optimal models and generating actionable insights to support data-driven educational solutions.

Data Scientist Intern

Stack Queue *Aug 2024 – Sep 2024*

- **Performed** comprehensive exploratory data analysis and time-series modeling on European ski-resort tourism datasets, uncovering seasonal trends and operational patterns to support strategic resource planning.
- **Developed** executive-level dashboards and analytical visualizations using Matplotlib and Seaborn, translating complex datasets into clear, actionable business insights for decision-makers.
- **Applied** statistical analysis, feature exploration, and trend identification techniques to evaluate tourism behavior, producing data-driven recommendations that improved forecasting accuracy and operational planning.

KEY PROJECTS

AI-Powered Financial Assistant — Multi-Agent GenAI System

Python • Google Gemini • LangChain • Model Context Protocol (MCP) • Flask • REST APIs

- Architected a multi-agent financial intelligence platform integrating Google Gemini, LangChain, and Model Context Protocol (MCP) to automate portfolio analysis, transaction monitoring, and personalized financial recommendations.
- Designed intelligent AI agents capable of contextual reasoning, investment analysis, and wealth projection through modular workflows, enabling adaptive decision-making across diverse financial scenarios.
- Developed a secure conversational interface using Flask and REST APIs, allowing users to interact with AI-powered financial insights through natural language while supporting scalable backend integration.

Real-Time Face & Hand Gesture Detection

Python • OpenCV • MediaPipe • Computer Vision • Landmark Detection

- Engineered a real-time computer vision application using MediaPipe landmark detection and OpenCV to simultaneously analyze facial expressions and hand gestures with live visual feedback.
- Implemented multi-modal gesture recognition for eye-state detection, smile recognition, and finger-count estimation, enabling intuitive human-computer interaction through real-time visual analytics.
- Optimized the vision processing pipeline for sub-100 ms inference latency on standard webcam hardware, delivering responsive real-time performance without dedicated GPU acceleration.

Student Malpractice Detection System — Real-Time AI Surveillance

Python • YOLOv8 • OpenCV • Flask • Computer Vision

- Developed an AI-powered examination monitoring system using a custom-trained YOLOv8 model to detect suspicious activities under diverse lighting and environmental conditions with robust real-time performance.
- Designed a Flask-based monitoring dashboard providing live detection visualization, automated alerts, and behavioral analytics to support intelligent examination surveillance.
- Applied custom dataset preparation, data augmentation, and model optimization techniques to improve detection accuracy, generalization, and real-time inference efficiency.

AI-Powered Plant Disease Detection

Python • TensorFlow • CNN • Transfer Learning • Flask • Image Classification

- Architected an AI-powered plant disease diagnosis system leveraging transfer learning and convolutional neural networks (CNNs) to accurately classify plant leaf diseases from real-world agricultural images.
- Integrated an intelligent treatment recommendation module delivering disease-specific guidance, enabling timely and informed decision-making for agricultural disease management.
- Deployed the complete solution as a Flask-based web application, providing an accessible AI-assisted diagnostic platform for farmers and non-technical users.

EDUCATION

B.Tech — Artificial Intelligence & Data Science **CGPA: 7.4 / 10 (up to Sem 7)**

Dhirajlal Gandhi College of Technology, Salem, Tamil Nadu

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Specialization – LIVEWIRE
- Master in Data Analysis and Analytics – Udemy
- AI for Bharat – Certificate of Recognition – Generative AI Content Summarizer using Amazon Bedrock | AWS × Hack2Skill
- Salem District-Level Forest Fire Prevention Hackathon, Salem (2nd Place)
- Google Agentic AI Day 2025 Hackathon (Shortlisted on National Level)
- Agentic Website Track, Bengaluru AI Hack Day (Participated)
- National Hackathon on Human-Wildlife Coexistence, Ministry of Environment, Government of India & Wildlife Institute of India, Dehradun (Participated)

LANGUAGES

Tamil • English